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Toner container and image forming apparatus including the toner container

Abstract

The toner container includes a container body, a transmission gear, a cap member, and a seal member. The transmission gear is fitted to the container body under a circumferential-movement restricted condition, and transmits rotation driving force to the container body. The cap member covers an opening of the container body, and retains a rotation-halted state while the container body and the transmission gear are rotating. The seal member is placed between the container body and the cap member. The transmission gear has a large-diameter portion larger in outer diameter than the other portions. The cap member includes a body portion, and a cylindrical-shaped retaining portion protruded from the body portion in a rotational-axis direction. A fore end portion of the retaining portion is in contact with a step gap portion on one side of the large-diameter portion closer to the cap member.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based on and claims the benefit of priority from Japanese Patent Application No. 2023-122339 filed on Jul. 27, 2023, the contents of which are hereby incorporated by reference.

BACKGROUND

(2) The present disclosure relates to a toner container being for use of containing toner and to be fitted to an image forming apparatus using electrophotographic process, and further relates to an image forming apparatus including the toner container.

(3) In an image forming apparatus of the electrophotographic process, toner is fed from a developing unit to an electrostatic latent image formed on a surface of an image carrier (e.g., photosensitive drum), allowing development process to be fulfilled. Toner to be used for such development process is fed from a toner container to the developing unit. The toner container is, for example, equipped with a container body that rotates while containing toner.

(4) With a toner container of such a configuration as described above, there are some cases where an opening of the container body is covered with a cover which does not rotate during the rotation

of the container body. In such a case, toner leakage is more likely to occur from between the container body and the cover, as compared with cases in which the cover rotates integrally with the container body during the rotation of the container body.

SUMMARY

(5) A toner container according to one aspect of the present disclosure includes a container body, a transmission gear, a cap member, and a seal member, the toner container being fittable to and removable from an image forming apparatus. The container body, being cylindrical-shaped, rotates while containing toner. The transmission gear, being cylindrical-shaped, is fitted to a one-end-side outer circumferential surface of the container body under a circumferential-movement restricted condition, and transmits rotation driving force to the container body. The cap member covers an opening provided at one end portion of the container body in its rotational-axis direction, and retains a rotation-halted state while the container body and the transmission gear are rotating. The seal member is placed between the container body and the cap member, and at least partly formed from an elastic material. The transmission gear has a large-diameter portion larger in outer diameter than the other portions. The cap member includes a body portion having a communicating portion communicatable with interior of the container body via the opening, a retaining portion, being cylindrical-shaped, which is protruded from the body portion toward the container body in the rotational-axis direction, and which retains part of both the container body and the transmission gear in the rotational-axis direction, and an engaging portion which is formed on an inner circumferential surface of the retaining portion, and which makes both a flange portion formed on a circumferential edge portion of the opening and the seal member pinched between the engaging portion and the body portion in the rotational-axis direction. A fore end portion of the retaining portion is in contact with a step gap portion on one side of the large-diameter portion closer to the cap member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram showing an internal configuration of an image forming apparatus in which toner containers according to an embodiment of the present disclosure are mounted;
- (2) FIG. 2 is an enlarged view of around an image forming part of FIG. 1;
- (3) FIG. 3 is a perspective view showing toner containers and container fitting portions in the embodiment;
- (4) FIG. 4 is a perspective view of a toner container according to the embodiment, as viewed from an upstream side in its insertion direction;
- (5) FIG. 5 is an exploded perspective view of a downstream-side end portion of the toner container according to the embodiment, as viewed in its insertion direction;
- (6) FIG. 6 is a sectional view of a downstream-side end portion of the toner container according to the embodiment, as viewed in its insertion direction and as cut along its axial direction;
- (7) FIG. 7 is an exploded view showing a layer structure of a seal member;
- (8) FIG. 8 is a sectional view of a downstream-side end portion of the toner container, as viewed in its insertion direction and as cut along its axial direction, where the toner container has been fitted to the container fitting portion;
- (9) FIG. 9 is an enlarged view of a contact portion between a transmission gear and a retaining portion of a cap member in FIG. 8; and
- (10) FIG. 10 is a side view schematically showing a connecting portion between a container body and the cap member in a toner container according to a modification of the embodiment.

DETAILED DESCRIPTION

- (11) Hereinafter, an embodiment of the present disclosure will be described with reference to the

accompanying drawings. FIG. 1 is a schematic diagram showing a configuration of an image forming apparatus **100** in which toner containers **4a** to **4d** according to an embodiment of the disclosure are mounted. FIG. 2 is an enlarged view of around an image forming part Pa of FIG. 1.

(12) The image forming apparatus **100** shown in FIG. 1 is a so-called tandem-type color printer, having a configuration described below. In a housing of the image forming apparatus **100**, four image forming parts Pa, Pb, Pc, Pd are disposed in this order starting from an upstream side (left side in FIG. 1) in a conveyance direction. These image forming parts Pa to Pd, which are provided in correspondence to images of different four colors (yellow, cyan, magenta and black), form images of yellow, cyan, magenta and black successively each through processes of charging, exposure to light, development and transfer.

(13) These image forming parts Pa, Pb, Pc, Pd are equipped with photosensitive drums **1a**, **1b**, **1c**, **1d**, respectively, for carrying individual-color visible images (toner images). Further, an intermediate transfer belt **8** that turns counterclockwise in FIG. 1 is provided in adjacency to the image forming parts Pa to Pd. Toner images formed on these photosensitive drums **1a** to **1d** are transferred successively onto the intermediate transfer belt **8** that is moving while keeping in contact with the photosensitive drums **1a** to **1d**, and thereafter transferred at one time onto a sheet S as an example of a recording medium in a secondary transfer roller **9**. Further, after fixation of the toner images on the sheet S in a fixing unit **13**, the sheet S is discharged out of the housing of the image forming apparatus **100**. While the photosensitive drums **1a** to **1d** are rotated clockwise in FIG. 1, image forming process for the individual photosensitive drums **1a** to **1d** is executed.

(14) The sheet S, to which toner images are to be transferred, is contained in a sheet cassette **16** placed at a lower portion in the housing of the image forming apparatus **100**. The sheet S is conveyed via a feed roller **12a** and a registration roller pair **12b** to the secondary transfer roller **9**. The intermediate transfer belt **8** is given mostly by using a seamless belt.

(15) Next, the image forming parts Pa to Pd will be described. Whereas the image forming part Pa will be described in detail below, the image forming parts Pb to Pd are basically similar in configuration thereto, so their description being omitted. As shown in FIG. 2, around the photosensitive drum **1a**, disposed along a drum rotational direction (clockwise direction in FIG. 2) are a charging unit **2a**, a developing unit **3a**, and a cleaning unit **7a**, where a primary transfer roller **6a** is placed with the intermediate transfer belt **8** pinched against the photosensitive drum **1a**. Also, a belt cleaning unit **19** opposed to a tension roller **11** with the intermediate transfer belt **8** pinched therebetween is placed on an upstream side of the photosensitive drum **1a** in a turning direction of the intermediate transfer belt **8**.

(16) Next, image formation procedure in the image forming apparatus **100** will be described. Upon input of a start for image formation from a user, the photosensitive drums **1a** to **1d** first start to be rotated by a main motor (not shown), and surfaces of the photosensitive drums **1a** to **1d** are uniformly electrically charged by charging rollers **20** of the charging units **2a** to **2d**, respectively. Subsequently, the surfaces of the photosensitive drums **1a** to **1d** are illuminated with optical beams (laser light) emitted from an exposure unit **5**, so that electrostatic latent images corresponding to image signals are formed on the photosensitive drums **1a** to **1d**, respectively.

(17) The developing units **3a** to **3d** are filled to specified quantities with individual-color toner of yellow, cyan, magenta and black, respectively. It is noted that when a ratio of toner contained in two-component developer filled in the developing units **3a** to **3d** has become lower than a predetermined value due to later-described formation of toner images, the developing units **3a** to **3d** are refilled with toner from the toner containers **4a** to **4d**, respectively. This toner in the developer is fed, by developing rollers **21** of the developing units **3a** to **3d**, onto the photosensitive drums **1a** to **1d**, respectively, so as to be electrostatically deposited thereon. As a result of this, a toner image corresponding to an electrostatic latent image formed by exposure from the exposure unit **5** is formed.

(18) Then, by the primary transfer rollers **6a** to **6d**, electric fields are given at a specified transfer

voltage to between the primary transfer rollers **6a** to **6d** and the photosensitive drums **1a** to **1d**, respectively, so that toner images of yellow, cyan, magenta and black on the photosensitive drums **1a** to **1d** are primarily transferred onto the intermediate transfer belt **8**. These images of four colors are formed with a specified positional relationship which is predetermined for specified full-color image formation. Thereafter, in preparation for subsequent formation of new electrostatic latent images, toner remaining on the surfaces of the photosensitive drums **1a** to **1d** is eliminated by cleaning blades **22** and rubbing rollers **23** of cleaning units **7a** to **7d**, respectively.

(19) As the intermediate transfer belt **8** has started to turn counterclockwise along with rotation of the driving roller **10** by a belt driving motor (not shown), the sheet **S** is conveyed at a specified timing from the registration roller pair **12b** to a secondary transfer roller **9** provided in adjacency to the intermediate transfer belt **8**, followed by transfer of the full-color image. The sheet **S**, to which the toner image has been transferred, is conveyed to the fixing unit **13**. Toner remaining on the surface of the intermediate transfer belt **8** is eliminated by the belt cleaning unit **19**.

(20) The sheet **S** conveyed to the fixing unit **13** is heated and pressurized by a fixing roller pair **13a** so that the toner image is fixed on the surface of the sheet **S**, with a specified full-color image formed. The sheet **S** with the full-color image formed thereon is assorted in conveyance direction by a branching part **14** that branches into a plurality of directions, and the sheet **S** is discharged, as it is (or after fed to a double-sided conveyance path **18** and subjected to double-sided printing), to a discharge tray **17** by a discharge roller pair **15**.

(21) Next, the toner containers **4a** to **4d** will be described. FIG. **3** is a perspective view showing the toner containers **4a** to **4d** and container fitting portions **30a** to **30d**. In FIG. **3**, only the toner container **4d** is shown as it is halfway on its fitting to the container fitting portion **30d** while the toner containers **4a** to **4c** are shown as they have completely been fitted to the container fitting portions **30a** to **30c**, respectively.

(22) As shown in FIG. **3**, the toner containers **4a** to **4d** are fitted to the container fitting portions **30a** to **30d**, respectively, along their insertion direction headed from fore side (right near side in FIG. **3**) toward rear side (left far side in FIG. **3**) of the image forming apparatus **100**. The container fitting portions **30a** to **30d** are arrayed in a left/right direction. The container fitting portions **30a** to **30d** are each formed into a cylindrical shape extending in a longitudinal (front/rear) direction. An outer diameter of the toner container **4d** in which black toner is contained is larger than an outer diameter of the toner containers **4a** to **4c** in which toner of yellow, cyan and magenta, respectively, is contained. An inner diameter of the container fitting portion **30d** is larger than an inner diameter of the container fitting portions **30a** to **30c**.

(23) A toner refill port **303** (see FIG. **8**) is provided at a lower portion of each of the container fitting portions **30a** to **30d** and on a downstream side of the toner-container insertion direction. The toner refill port **303** is communicated with interior of each of the developing units **3a** to **3d** of the image forming parts **Pa** to **Pd**, respectively, via a refill tube **304** (see FIG. **8**) extending in an up/down direction.

(24) FIG. **4** is a perspective view of a toner container **4d** according to the embodiment, as viewed from an upstream side in its insertion direction. FIG. **5** is an exploded perspective view of a downstream-side (right far side of FIG. **4**) end portion of the toner container **4d** according to the embodiment, as viewed in its insertion direction. FIG. **6** is a sectional view of a downstream-side end portion of the toner container **4d** according to the embodiment, as viewed in its insertion direction and as cut along its axial direction. Although the toner container **4d** for containing toner of black will be described below, the toner containers **4a** to **4c** for containing toner of yellow, cyan and magenta, respectively, are similar in configuration to the toner container **4d**.

(25) As shown in FIGS. **4** and **5**, the toner container **4d** includes a container body **40**, a transmission gear **41** fitted to a rear portion of the container body **40**, a cap member **42** provided on a rear side of the transmission gear **41**, and a seal member **43** housed inside the cap member **42**.

(26) The container body **40** is formed into a cylindrical shape extending along the insertion

direction (arrow A direction) relative to the container fitting portion **30d**. In this embodiment, a front/rear direction of the image forming apparatus **100** is a longitudinal direction of the container body **40**. The container body **40** is made up so as to be rotatable about a rotational axis Y extending in the longitudinal direction. Toner of black (not shown) is contained inside the container body **40**. The toner contained inside the container body **40** is conveyed from front side toward rear side (from upstream side toward downstream side of insertion direction) of the image forming apparatus **100**.

(27) The container body **40** has a handle portion **401**, a large-diameter cylindrical portion **402** provided downstream of the handle portion **401** in a toner conveyance direction (arrow A direction), a middle-diameter cylindrical portion **403** provided downstream of the large-diameter cylindrical portion **402**, and a small-diameter cylindrical portion **404** provided downstream of the middle-diameter cylindrical portion **403**.

(28) The handle portion **401** is provided at an upstream-side end portion of the container body **40** in its insertion direction. An inner space of the handle portion **401** is communicated with an inner space of the large-diameter cylindrical portion **402**. The large-diameter cylindrical portion **402**, the middle-diameter cylindrical portion **403**, and the small-diameter cylindrical portion **404** are formed each into a cylindrical shape centered on the rotational axis Y. The middle-diameter cylindrical portion **403** is provided between the large-diameter cylindrical portion **402** and the small-diameter cylindrical portion **404**. The middle-diameter cylindrical portion **403** is smaller in outer diameter than the large-diameter cylindrical portion **402** and larger in outer diameter than the small-diameter cylindrical portion **404**.

(29) On an outer circumferential surface of the container body **40**, a first step-gap portion **405** having a generally annular shape is formed between the large-diameter cylindrical portion **402** and the middle-diameter cylindrical portion **403**, and a second step-gap portion **406** having a spiral shape is formed between the middle-diameter cylindrical portion **403** and the small-diameter cylindrical portion **404**. The first step-gap portion **405** has a step gap (difference in outer diameter between the large-diameter cylindrical portion **402** and the middle-diameter cylindrical portion **403**) smaller than a step gap (difference in outer diameter between the middle-diameter cylindrical portion **403** and the small-diameter cylindrical portion **404**) of the second step-gap portion **406**.

(30) One spiral conveyance rib **407** is continuously provided on inner circumferential surfaces of the large-diameter cylindrical portion **402**, the middle-diameter cylindrical portion **403**, and the small-diameter cylindrical portion **404** in the container body **40**. The conveyance rib **407** projects, in a generally V shape in a sectional view, from the inner circumferential surface of the container body **40** toward radially inner side.

(31) On an outer circumferential surface of the small-diameter cylindrical portion **404**, a pair of first engaging protrusions **50** (rotation restricting portions) are provided. One of the paired first engaging protrusions **50** is placed so as not to overlap with the conveyance rib **407**. The other of the paired first engaging protrusions **50** is placed so as to overlap with the conveyance rib **407**. The pair of first engaging protrusions **50** are provided at positions of point symmetry relative to the rotational axis Y.

(32) On the outer circumferential surface of the small-diameter cylindrical portion **404**, a pair of second engaging protrusions **51** (movement restricting portions) are provided. The second engaging protrusions **51** are placed so as neither to overlap with the conveyance rib **407**. The second engaging protrusions **51** are placed with a circumferential-direction positional shift of 90 degrees relative to the first engaging protrusions **50**, respectively. That is, the pair of first engaging protrusions **50** and the pair of second engaging protrusions **51** are placed so as not to overlap with each other, respectively, in terms of their circumferential-direction positions.

(33) A circular-shaped opening **52** is provided at a downstream-side end portion of the small-diameter cylindrical portion **404** in the insertion direction. On a circumferential edge portion of the opening **52**, a ring-shaped flange portion **53** is provided so as to protrude radially outward.

(34) The transmission gear **41** is fitted to the outer circumferential surface of the small-diameter cylindrical portion **404** in the container body **40**. The transmission gear **41** is formed into a cylindrical shape centered on the rotational axis Y. The transmission gear **41** has a large-diameter portion **411**, a middle-diameter portion **412** provided downstream of the large-diameter portion **411** in the toner conveyance direction, and a small-diameter portion **413** provided downstream of the middle-diameter portion **412**. In the transmission gear **41**, its outer diameter decreases more and more in order of the large-diameter portion **411** to the middle-diameter portion **412** to the small-diameter portion **413**.

(35) On an outer circumferential surface of the large-diameter portion **411**, gear teeth **55** are provided. A diameter of the large-diameter portion **411** including the gear teeth **55** is smaller than the outer diameters of the large-diameter cylindrical portion **402** and the middle-diameter cylindrical portion **403** in the container body **40**, and meanwhile larger than the outer diameter of the small-diameter cylindrical portion **404**. An upstream-side end portion of the large-diameter portion **411** in the insertion direction is opposed to the second engaging protrusions **51** of the container body **40**.

(36) Over whole inner circumferential surfaces of the large-diameter portion **411** and the middle-diameter portion **412** in the transmission gear **41**, a plurality of engaging ribs **56** extending along the insertion direction are provided. The plurality of engaging ribs **56** are provided with specified circumferential intervals therebetween (e.g., **10** to **20** ribs at equal intervals).

(37) The one pair of first engaging protrusions **50** are inserted between two sets of engaging ribs **56**, respectively, out of the plurality of engaging ribs **56**. As a result of this, rotation of the transmission gear **41** relative to the container body **40** is restricted. It is noted that the engaging ribs **56** other than the above-mentioned two sets of engaging ribs **56** (engaging ribs **56** into which no first engaging protrusion **50** is inserted) function as reinforcers for reinforcing the transmission gear **41**.

(38) The cap member **42** covers the opening **52** of the container body **40**. The cap member **42** has a body portion **421**, a retaining portion **422** extending from the body portion **421** toward insertion-direction upstream side, an engaging cylindrical portion **423** protruding from the body portion **421** toward insertion-direction downstream side, a cover-side restricting portion **424** provided on an upper side of the engaging cylindrical portion **423**, and a cover-side shutter **425** provided on a lower side of the body portion **421**, the retaining portion **422** and the engaging cylindrical portion **423**.

(39) Inside the body portion **421**, a communicating portion **60** is provided so as to be communicatable with a toner containing space of the container body **40** via the opening **52** of the container body **40**. At a lower end of the communicating portion **60**, a toner discharge port **61** is provided.

(40) The retaining portion **422** is formed into a generally cylindrical shape centered on the rotational axis Y so as to protrude from the body portion **421** toward the container body **40** side. An inner circumferential surface of the retaining portion **422** is opposed to the outer circumferential surfaces of the middle-diameter portion **412** and the small-diameter portion **413** in the transmission gear **41**. On the inner circumferential surface of the retaining portion **422** of the cap member **42**, a total of four engaging portions **62** are provided. Each engaging portion **62** holds the flange portion **53** (downstream-side end portion in insertion direction) and the seal member **43** pinched in a clearance D against the body portion **421**. The clearance D is narrower than a sum of thicknesses of the flange portion **53** and the seal member **43**. Therefore, the seal member **43** is compressed to a specified thickness, so that scalability between the container body **40** and the cap member **42** is maintained.

(41) The retaining portion **422** holds the transmission gear **41** pinched against the second engaging protrusions **51** of the container body **40**. As a result, insertion-direction movement of the transmission gear **41** relative to the container body **40** is restricted while the transmission gear **41** is

prevented from dropping off from the container body **40**.

(42) The seal member **43** is formed into a ring shape and placed radially inside the retaining portion **422** of the cap member **42**. The seal member **43** is placed, in a compressed state, between the flange portion **53** of the container body **40** and the body portion **421** of the cap member **42**.

(43) The seal member **43** is fixed to a surface (upstream-side surface in insertion direction) of the body portion **421** of the cap member **42**, the surface being opposed to the flange portion **53**. In addition, although the seal member **43** may be fixed to the flange portion **53** of the container body **40**, yet there is a possibility that with the seal member **43** fixed to the flange portion **53**, the seal member **43** may be damaged by making contact with the retaining portion **422** during fitting of the cap member **42** to the container body **40**. Therefore, in terms of assemblability of the toner container **4d**, it is preferable to fix the seal member **43** to the body portion **421** of the cap member **42**.

(44) FIG. 7 is an exploded view showing a layer structure of the seal member **43**. The seal member **43** includes a first seal layer **431**, a first adhesion layer **432**, a second seal layer **433**, and a second adhesion layer **434**. The first seal layer **431**, forming a topmost layer of the seal member **43**, is opposed to the flange portion **53** of the container body **40**. The first seal layer **431** is formed from, for example, urethane foam, which is an elastic material. The first adhesion layer **432** is stacked on a back side (downstream side in insertion direction) of the first seal layer **431**. The first adhesion layer **432** is formed of, for example, double-sided tape to adhesively bond the first seal layer **431** and the second seal layer **433** together.

(45) The second seal layer **433** is stacked on a back side (downstream side in insertion direction) of the first adhesion layer **432**. The second seal layer **433** has a thickness occupying 50% or more of a whole thickness of the seal member **43**. The second seal layer **433** is formed from, for example, urethane foam, which is an elastic material. The second adhesion layer **434** is stacked on a back side (downstream side in insertion direction) of the second seal layer **433**. The second adhesion layer **434** is formed of, for example, double-sided adhesive tape to adhesively bond the second seal layer **433** and the body portion **421** of the cap member **42** together.

(46) Next, an operation of fitting the toner containers **4a** to **4d** to the container fitting portions **30a** to **30d** will be described. In addition, because the fitting operation is similar among the toner containers **4a** to **4d**, only the operation of fitting the toner container **4d** to the container fitting portion **30d** will be described below, and description of operations of fitting the toner containers **4a** to **4c** to the container fitting portions **30a** to **30c**, respectively, will be omitted.

(47) In course of fitting the toner container **4d** to the container fitting portion **30d**, as shown in FIG. 2, the toner container **4d** is inserted into the container fitting portion **30d**, starting with a downstream-side end portion (cap member **42** side end portion) of the toner container **4d** in its insertion direction.

(48) FIG. 8 is a sectional view of a downstream-side end portion of the toner container **4d**, as viewed in its insertion direction and as cut along its axial direction, where the toner container **4d** has been fitted to the container fitting portion **30d**. When the toner container **4d** has been fitted to the container fitting portion **30d**, the body portion **421** of the cap member **42** presses a boss **302** of a fitting portion-side shutter **301** of the container fitting portion **30d** while the engaging cylindrical portion **423** of the cap member **42** in the toner container **4d** keeps engaged with the boss **302**, as shown in FIG. 8. As a result of this, the fitting portion-side shutter **301** is moved from a closure position to an open position, allowing the toner refill port **303** of the container fitting portion **30d** to be opened.

(49) Also, when the toner container **4d** has been fitted to the container fitting portion **30d**, the cover-side shutter **425** provided in the cap member **42** is moved from a closure position to an open position, allowing the toner discharge port **61** of the body portion **421** in the cap member **42** to be opened. As a result, the communicating portion **60** of the body portion **421** is communicated with the refill tube **304**.

(50) Also, when the toner container **4d** has been fitted to the container fitting portion **30d** as described above, a fit-in protrusion **63** provided in the cover-side restricting portion **424** of the cap member **42** in the toner container **4d** is fitted into a fitting portion-side restricting portion **305** of the container fitting portion **30d**. As a result of this, the toner container **4d** is positioned relative to the container fitting portion **30d**.

(51) Next, an operation of refilling toner from the toner container **4d** to the developing unit **3d** of the image forming part Pd in the image forming apparatus **100** will be described. In addition, operations of refilling toner from the toner containers **4a** to **4c** to the developing units **3a** to **3c** of the image forming parts Pa to Pc, respectively, are similar to that of the toner container **4d**, and therefore their description is omitted.

(52) In the course of refilling toner from the toner container **4d** to the developing unit **3d** of the image forming part Pd, a toner refilling motor (not shown) is driven with the toner container **4d** fitted to the container fitting portion **30d** as shown in FIG. 7. As a result, rotation driving force from the toner refilling motor is transmitted via a gear train and a drive input gear (neither shown) to the transmission gear **41** of the toner container **4d**, causing the transmission gear **41** to be rotated.

(53) As the transmission gear **41** is rotated, the container body **40** of the toner container **4d** is resultantly rotated integrally with the transmission gear **41**. That is, the rotation driving force derived from the toner refilling motor is transmitted to the container body **40** by the transmission gear **41**.

(54) Meanwhile, the cap member **42** and the seal member **43** of the toner container **4d** keep in a rotation-halted state without being rotated during the rotation of the container body **40**. Therefore, the container body **40** is rotated relative to the cap member **42** and the seal member **43**.

(55) When the container body **40** is rotated, toner within the container body **40** is conveyed from upstream side toward downstream side of the toner conveyance direction by the conveyance rib **407** of the container body **40**, being discharged from the container body **40** via the opening **52**. The toner discharged from the container body **40** via the opening **52** is introduced to the communicating portion **60** of the body portion **421** in the cap member **42**. The toner introduced to the communicating portion **60** is discharged from the communicating portion **60** via the toner discharge port **61** of the body portion **421**. The toner discharged from the communicating portion **60** via the toner discharge port **61** is introduced to the refill tube **304** via the toner refill port **303** of the container fitting portion **30d**, being refilled from the refill tubes **304** into the developing unit **3d** of the image forming part Pd.

(56) As described above, with the toner containers **4a** to **4d** fitted to the container fitting portions **30a** to **30d**, respectively, only the flange portion **53** (downstream-side end portion in insertion direction) of the container body **40** is retained by the cap member **42**. With this configuration, there are some cases where the container body **40** is tilted downward relative to the cap member **42** due to weight of toner contained inside the container body **40** as well as self weight of the container body **40**. In such a state, when the container body **40** is rotated, a load applied to lower part of the seal member **43** becomes larger, leading to occurrence of partial wear or exhaustion of the seal member **43**.

(57) In the course of fitting the toner containers **4a** to **4d** to the container fitting portions **30a** to **30d**, respectively, when the container body **40** is pushed inward to an extent deeper than a specified position, the seal member **43** is excessively compressed. The seal member **43**, when stored in that state for a long period, would yield to exhaustion.

(58) Upon occurrence of wear or exhaustion of the seal member **43**, toner is more likely to leak from between the container body **40** and the cap member **42**. Also, since the seal member **43** becomes shorter in service life, there arises a need for replacing the seal member **43** on occasions when toner is filled, for reuse, into the used toner containers **4a** to **4d**, hence lowered reusability.

(59) FIG. 9 is an enlarged view of a contact portion between the transmission gear **41** and the retaining portion **422** of the cap member **42** in FIG. 8. In this embodiment, as shown in FIG. 9, the

retaining portion **422** of the cap member **42** is lengthened toward the container body **40** side (upstream side in insertion direction). In more detail, a fore end portion **422a** of the retaining portion **422** is set in contact with a step gap portion **414** between the large-diameter portion **411** and the middle-diameter portion **412** of the transmission gear **41**.

(60) With this configuration, the container body **40** and the cap member **42** are restricted in position in the longitudinal direction (rotational axis Y direction) by the contact between the fore end portion **422a** of the retaining portion **422** and the step gap portion **414** of the transmission gear **41**. Accordingly, even when the toner containers **4a** to **4d** are strongly pushed in to the container fitting portions **30a** to **30d**, respectively, the seal member **43** is kept from action of pressing force. That is, since the compressed state of the seal member **43** is maintained constant by the clearance D between the engaging portions **62** and the body portion **421**, exhaustion of the seal member **43** due to excessive compression can be suppressed.

(61) Further, the inner circumferential surface **422b** of the retaining portion **422** is set in contact with the outer circumferential surface of the middle-diameter portion **412** of the transmission gear **41**. With this configuration, portions of the container body **40** and the transmission gear **41** which range from the flange portion **53** to the step gap portion **414** of the transmission gear **41** are axially supported by the cap member **42**. As a result, an axial retaining length of the container body **40** and the transmission gear **41** becomes so large that a load with which the container body **40** tends to be tilted downward can be held not only by vicinities of the seal member **43** but also by axial diffusion of the load.

(62) Furthermore, the container body **40** and the cap member **42** are restricted in radial (in a direction perpendicular to the rotational axis Y) position by a contact between the inner circumferential surface **422b** of the retaining portion **422** and the middle-diameter portion **412** of the transmission gear **41**. In this connection, a tilt quantity of the container body **40** is determined by a vicinity (axial support portion) of the seal member **43** or the retaining portion **422** (longitudinal support portion), whichever it is the one of smaller radial clearance (backlash). Because, normally, fit-in between the inner circumferential surface **422b** of the retaining portion **422** and the middle-diameter portion **412** of the transmission gear **41** is capable of more easily obtaining dimensional accuracy, lengthening the retaining portion **422** allows further radial-position restricting effect to be obtained more easily. Preferably, a distance (axial support distance) ranging from the flange portion **53** to the fore end portion **422a** of the retaining portion **422** is set larger than a radius R of the opening **52**.

(63) As described above, according to this embodiment, tilt of the container body **40** can be restricted without applying any excessive load to the seal member **43**, so that toner leakage of the toner containers **4a** to **4d** can be suppressed. Further, wear or exhaustion of the seal member **43** can also be suppressed, leading to longer service life of the seal member **43**. Thus, there can be provided toner containers **4a** to **4d** that are capable of prolonged use and moreover repetitive use enabled by refilling of toner.

(64) Without being limited to the above-described embodiment, the present disclosure may be changed and modified in various ways unless those changes and modifications depart from the gist of the disclosure. For example, in the above embodiment, the large-diameter portion **411** of the transmission gear **41** is provided at an end portion on one side of the transmission gear **41** farther from the cap member **42** in the rotational-axis direction, and the gear teeth **55** are formed on the outer circumferential surface of the large-diameter portion **411**. However, this is not limitative, and the gear teeth **55** may also be formed at a portion other than the large-diameter portion **411** of the transmission gear **41**.

(65) For instance, like a modification shown in FIG. **10**, the large-diameter portion **411** may be provided at a generally central portion of the transmission gear **41** in the rotational-axis direction, and the gear teeth **55** may be formed at an end portion on one side of the transmission gear **41** farther from the cap member **42**. With this configuration as well, the retaining portion **422** of the

cap member **42** makes contact with the step gap portion **414** on one side (downstream side in insertion direction) of the large-diameter portion **411** closer to the cap member **42**, so that the container body **40** and the cap member **42** are restricted in terms of their longitudinal positions. Further, by the retaining portion **422** making contact with the outer circumferential surface of the middle-diameter portion **412** of the transmission gear **41**, the container body **40** and the cap member **42** are restricted in terms of their radial positions.

(66) The above embodiment has been described about the toner containers **4a** to **4d** which contain toner for refilling the developing units **3a** to **3d**, respectively, and which are mounted on the image forming apparatus **100** with the developing units **3a** to **3d** mounted thereon, the developing units **3a** to **3d** being of the two-component development method using two-component developer containing magnetic carrier and toner. However, the present disclosure is applicable similarly also to toner containers which contain toner for refilling developing units of the magnetic one-component development method using magnetic toner or developing units of the nonmagnetic one-component development method using nonmagnetic toner.

(67) Furthermore, the present disclosure is applicable to a variety of image forming apparatuses using toner containers, such as not only tandem-type color printers as shown in FIG. **1** but also color copiers or color multifunction peripherals as well as monochromatic printers or monochromatic copiers or monochromatic multifunction peripherals.

(68) This disclosure can be exploited for toner containers which contain toner therein. Exploiting the disclosure makes it possible to provide toner containers, as well as image forming apparatuses including such toner containers, in which wear or distortion of the seal member can be suppressed by reducing variations of pressing force acting on the seal member.

Claims

1. A toner container comprising: a container body, being cylindrical-shaped, which rotates while containing toner; a transmission gear, being cylindrical-shaped, which is fitted to a one-end-side outer circumferential surface of the container body under a circumferential-movement restricted condition, and which transmits rotation driving force to the container body; a cap member which covers an opening provided at one end portion of the container body in its rotational-axis direction, and which retains a rotation-halted state while the container body and the transmission gear are rotating; and a seal member which is placed between the container body and the cap member, and which is at least partly formed from an elastic material, the toner container being fittable to and removable from an image forming apparatus, wherein the transmission gear has a large-diameter portion larger in outer diameter than the other portions, and the cap member includes: a body portion having a communicating portion communicatable with interior of the container body via the opening; a retaining portion, being cylindrical-shaped, which is protruded from the body portion toward the container body in the rotational-axis direction, and which retains part of both the container body and the transmission gear in the rotational-axis direction; and an engaging portion which is formed on an inner circumferential surface of the retaining portion, and which makes both a flange portion formed on a circumferential edge portion of the opening and the seal member pinched between the engaging portion and the body portion in the rotational-axis direction, and a fore end portion of the retaining portion is in contact with a step gap portion on one side of the large-diameter portion closer to the cap member.
2. The toner container according to claim 1, wherein the inner circumferential surface of the retaining portion is in contact with an outer circumferential surface of the transmission gear on one side closer to the cap member than to the step gap portion.
3. The toner container according to claim 1, wherein a distance from the fore end portion of the retaining portion to the opening is larger than a radius of the opening.
4. The toner container according to claim 1, wherein the large-diameter portion is formed at an end

portion of the transmission gear on its one side farther from the cap member in the rotational-axis direction, and gear teeth are formed on an outer circumferential surface of the large-diameter portion.

5. An image forming apparatus comprising: the toner container according to claim 1; and a fitting portion to which the toner container is fitted.
